

Framework question	Trial answer for VELFAC 200 ENERGY case	Evidence basis	Risk classification	Confidence	Implication for the framework
Q1 What function must be preserved?	The window must provide one residential window opening with equivalent thermal, acoustic, safety and weather-protection performance over 50 years	Functional unit and lifecycle scenario comparison	Lifecycle verification	Medium	The case can only be compared fairly if repaired and replaced options preserve equivalent performance
Q2 Which components fail first?	Likely shorter-life components include hardware, opening mechanisms, handles, fittings, seals, gaskets, glazing unit edge seal and surface finish	Component map and expert interviews	Premature replacement risk	Medium	The framework should first identify which parts fail before the full frame
Q3 Can components be replaced independently?	Some components appear repairable or replaceable, including selected fittings, handles, hardware and possibly glazing-related components, but the full replaceability route is not completely confirmed from public information	VELFAC documentation and interview evidence	Implementation gap	Medium	Technical repairability should be treated as partly confirmed, not fully proven
Q4 Is repair practically enabled?	Maintenance and product documentation exist, but practical repair depends strongly on installation access, wall and window interface, service clearance and long-term spare-part availability	VELFAC case uncertainty and interviews	Implementation gap	Medium	The framework must check product design and building installation together
Q5 Which actors must support the strategy?	Manufacturer, supplier, installer, architect, client and maintenance provider all affect whether repair remains possible over time	Interview memo and risk pathways	Implementation gap	Medium	Repairability depends on responsibility across the supply chain, not only on the product
Q6 What trade-offs are introduced?	The wood-aluminium frame and triple glazing may support durability and performance, but high-performance construction may also involve material and complexity trade-offs that need verification	Lifecycle scenario comparison and assumptions table	Lifecycle trade-off	Medium	The framework should avoid assuming that higher performance or longer life is automatically better
Q7 Could rebound or displacement occur?	Possible risks include client preference for newer products, repeated service transport, difficult reuse due to custom dimensions and uncertain end-of-life recovery	Interviews, reuse exclusion and classified pathways	Potential behavioural rebound, burden shifting and implementation gap	Low to medium	The framework should classify several possible mechanisms rather than forcing all risks into rebound
Q8 Is lifecycle benefit plausible?	Component repair and replacement appear plausible if access, spare parts, maintenance and performance are maintained, but the benefit cannot be claimed quantitatively without stronger data	Transparent assumptions table and lifecycle scenario comparison	Lifecycle verification	Medium	The case supports a cautious conditional conclusion rather than a measured environmental saving